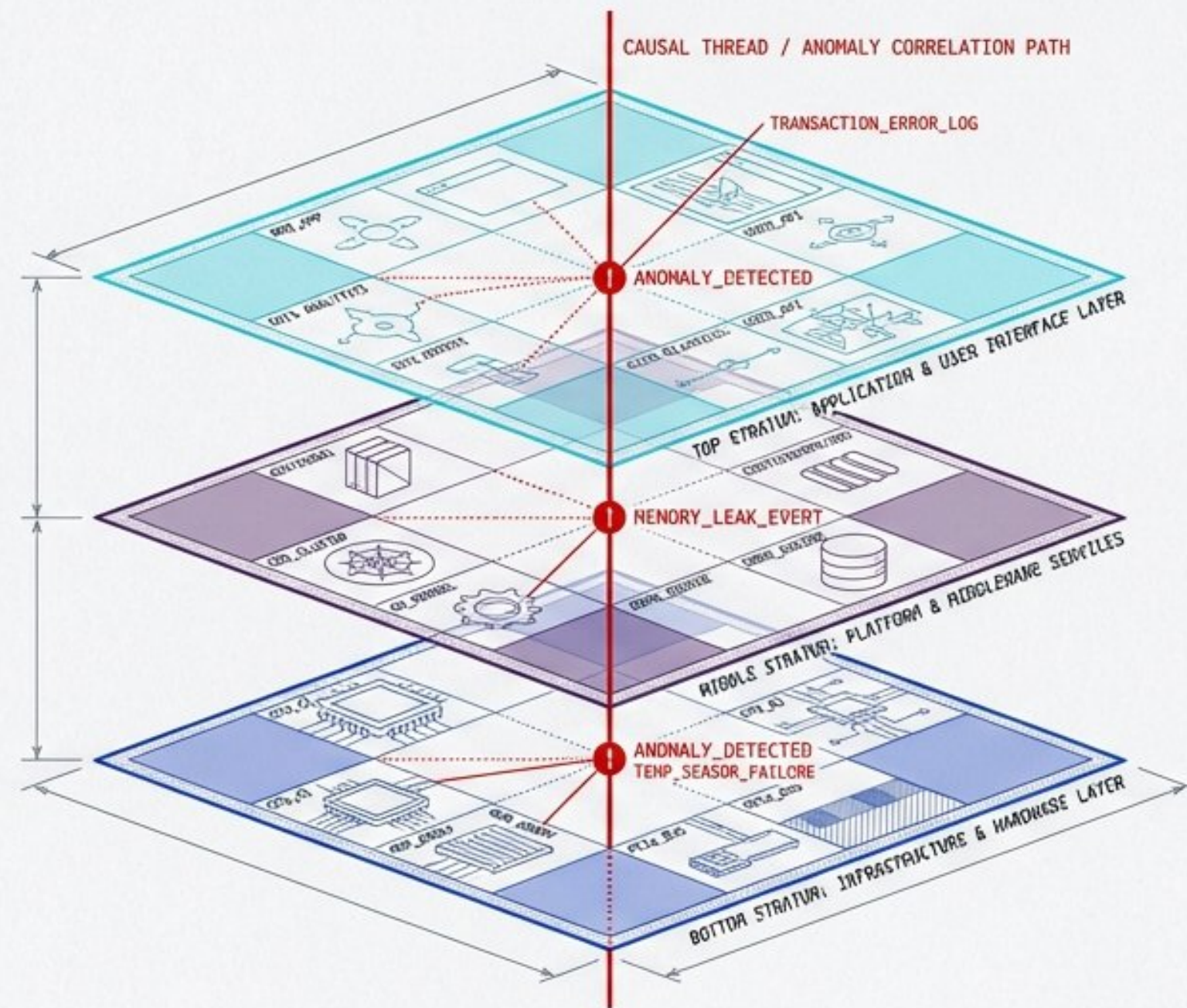


Autonomous Multi-Stratum Anomaly Correlation Engine

Solving the Cross-Stratum Blind Spot in Modern Computing Systems

Version 2.0 Specification Outline / Attorney-Client Privilege

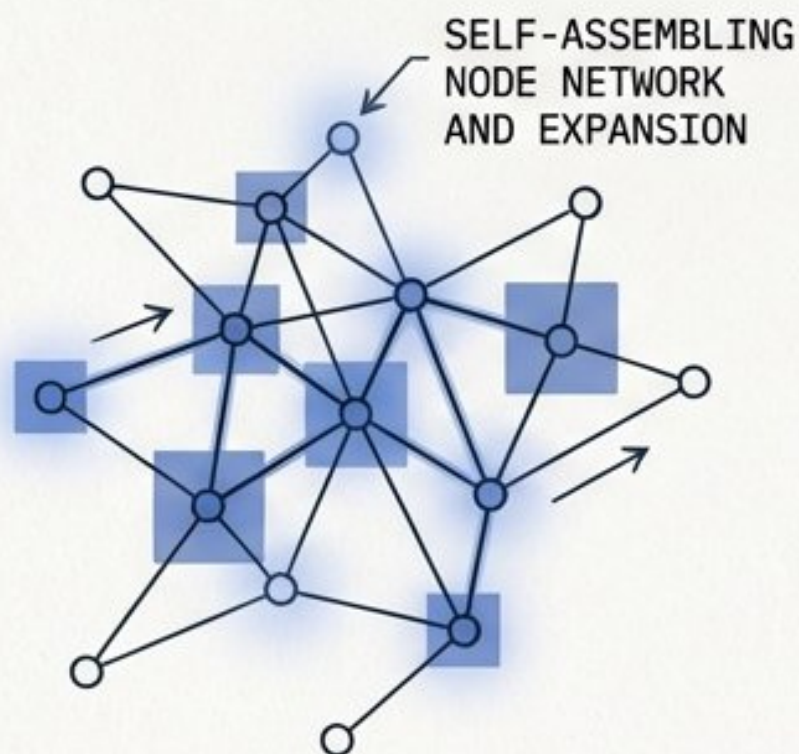


The AMACE Paradigm

AMACE autonomously ingests heterogeneous data, correlates anomalies across strata using information-theoretic dimensional pruning, and validates causation via counterfactual analysis—provably within sub-quadratic computational complexity.

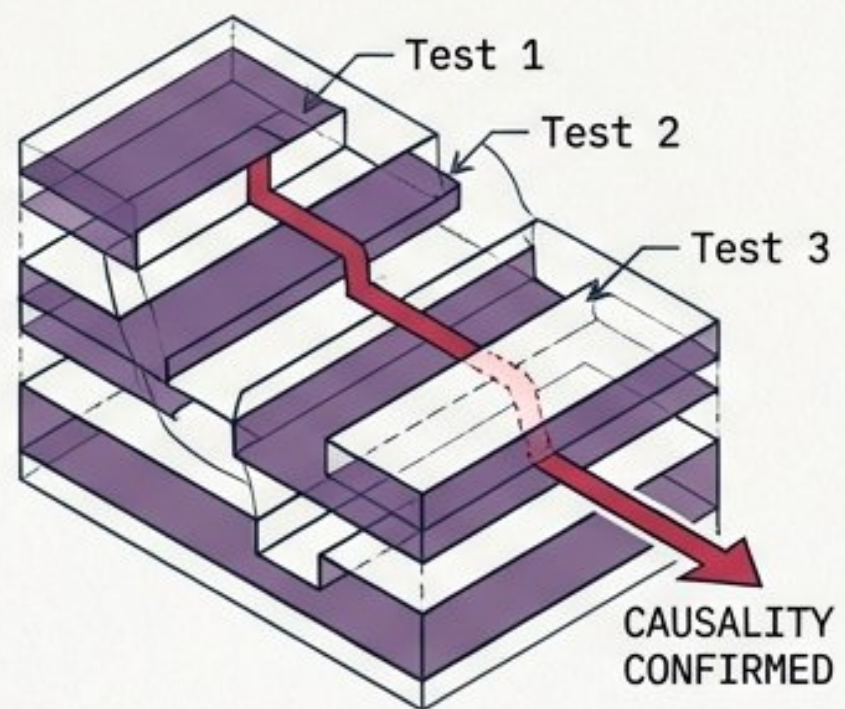
Zero-Configuration

No human setup, predefined rules, or domain-specific knowledge required.



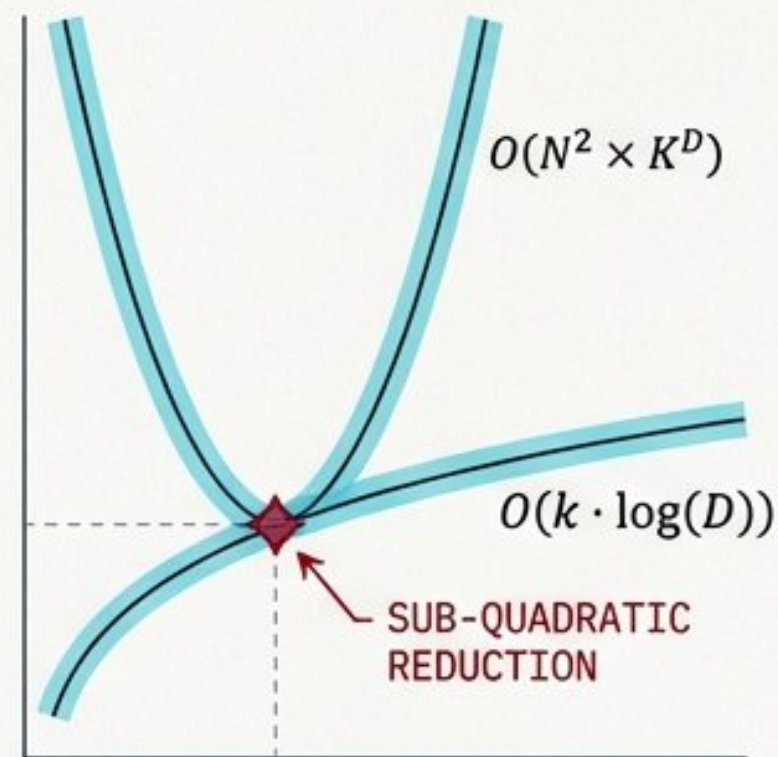
Provable Causality

Replaces correlation heuristics with formal three-test computational ablation.

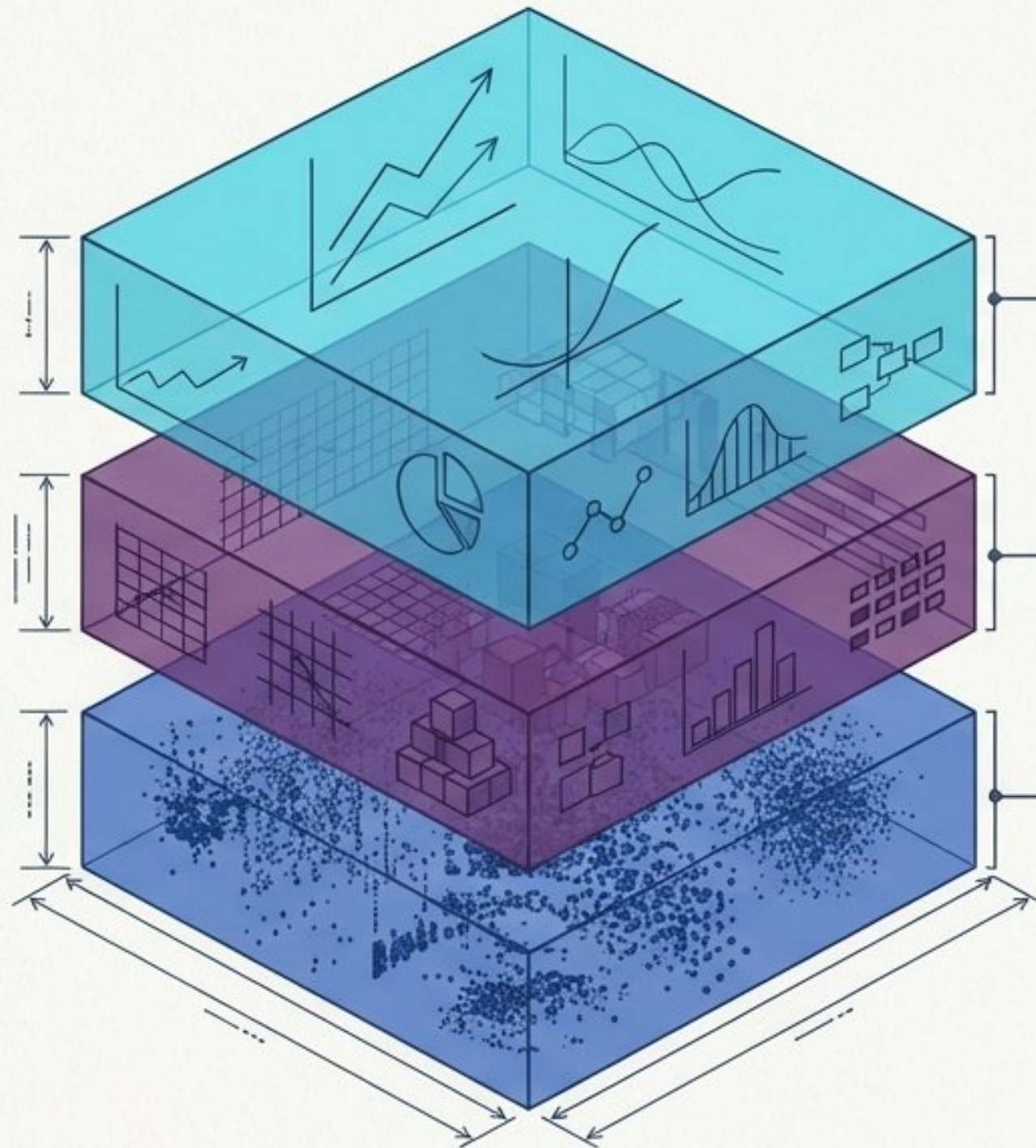


Sub-Quadratic Complexity

Exponentially reduces $O(N^2 \times K^D)$ search space to $O(k \cdot \log(D))$.



The Architecture of Modern Systems



S2: Inference Stratum

High-level predictive intelligence, ML model outputs, and structured business logic.

Low volume, highly structured.

S1: Representation Stratum

Computed statistical aggregations, derived metrics, and engineered knowledge.

Moderate volume, structured.

S0: Observation Stratum

Raw, unstructured event data (10^6 – 10^{12} events/day).

No semantic interpretation.

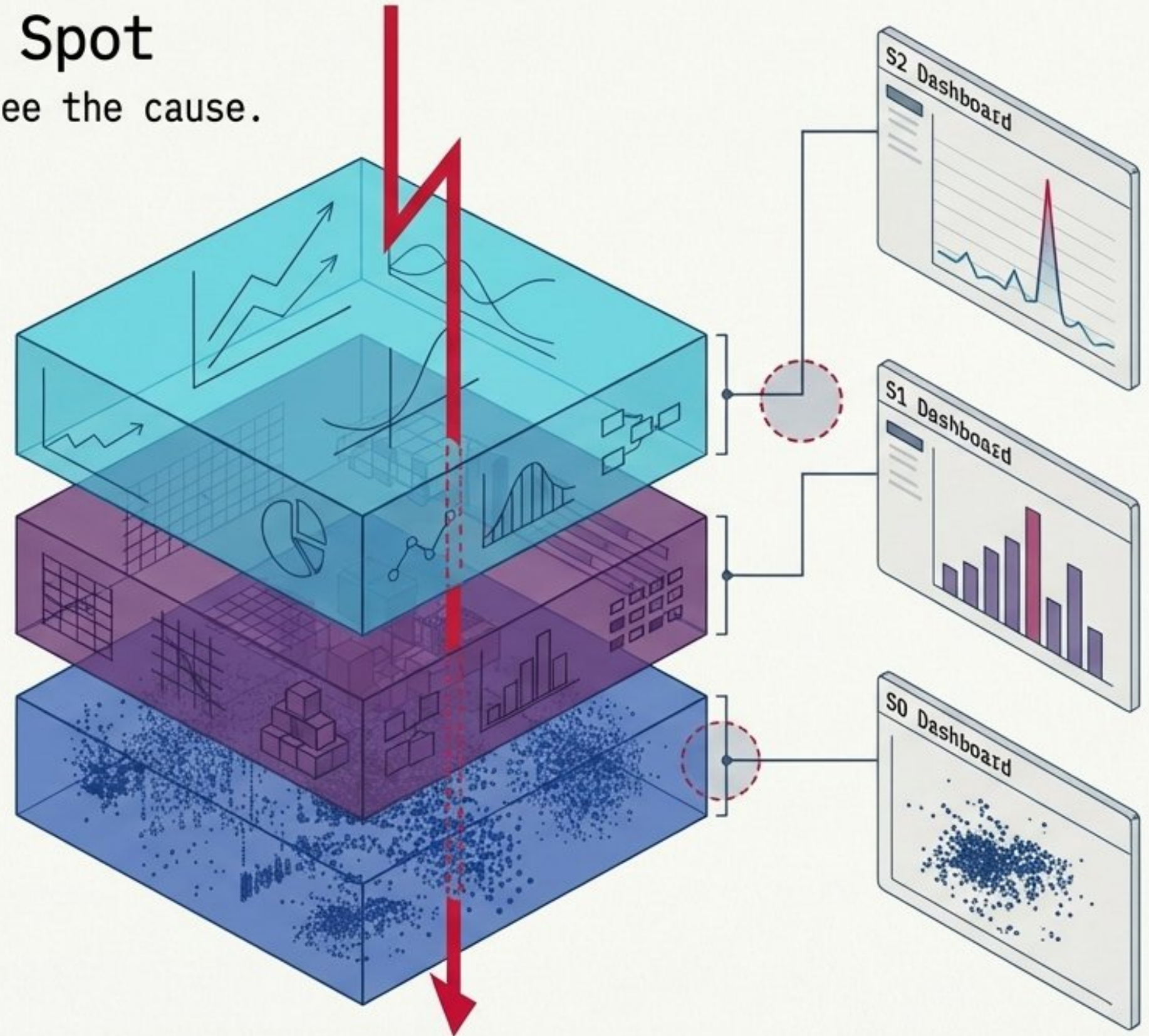
The Cross-Stratum Blind Spot

Each tool sees its own symptoms; none see the cause.

Existing anomaly detection tools are architecturally isolated. They operate within a single stratum.

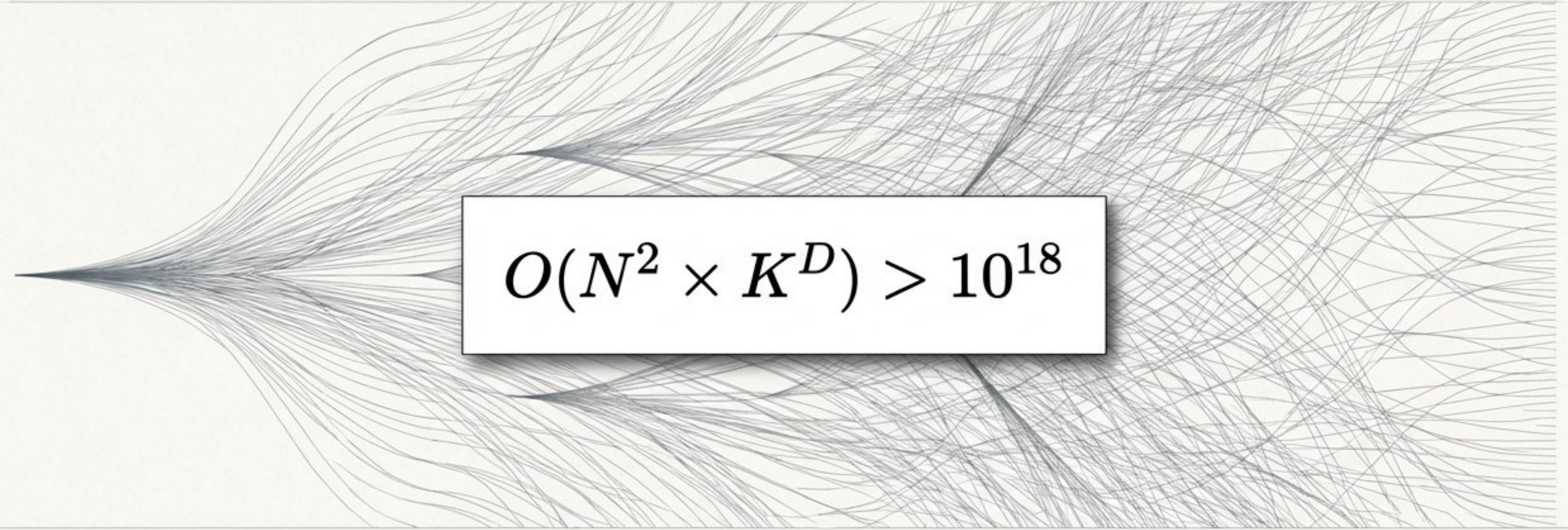
When a malfunction cascades vertically across S0, S1, and S2, each tool sees only local symptoms.

None can determine the shared causal origin.



The Computational Intractability of Causation

Cross-stratum correlation isn't just an engineering gap—it's computationally intractable. Until now.


$$O(N^2 \times K^D) > 10^{18}$$

Combinatorial Explosion

Given N anomalies across 3 strata, D dimensions, and K average cardinality, exhaustive correlation requires over 10^{18} evaluations.

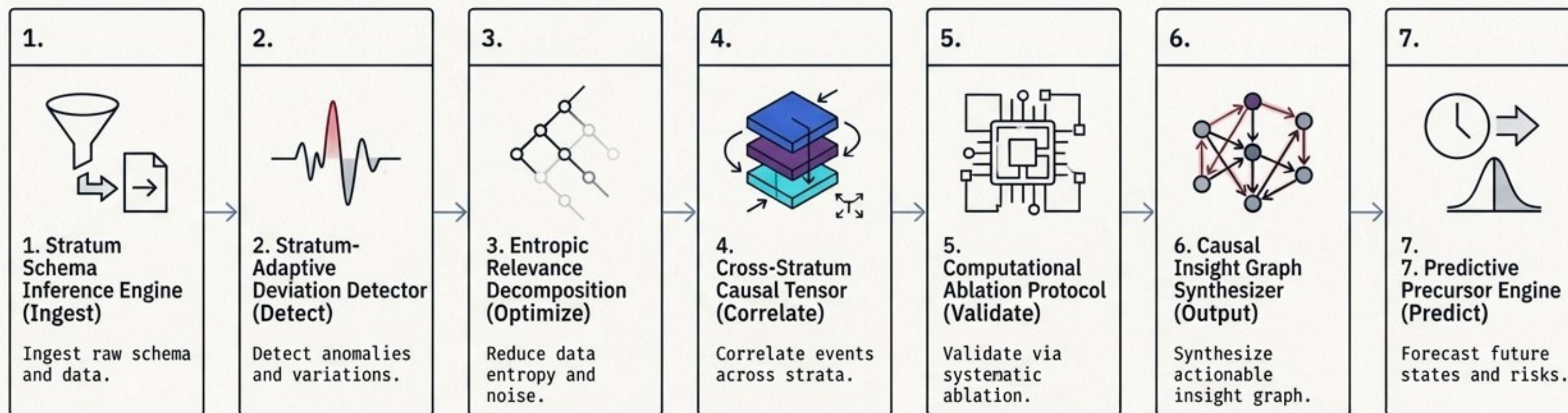
Heterogeneous Schema

Strata lack common keys.
Entity resolution is NP-hard.

Causal Asymmetry

Standard correlation is symmetric. It cannot separate cause from coincidence.

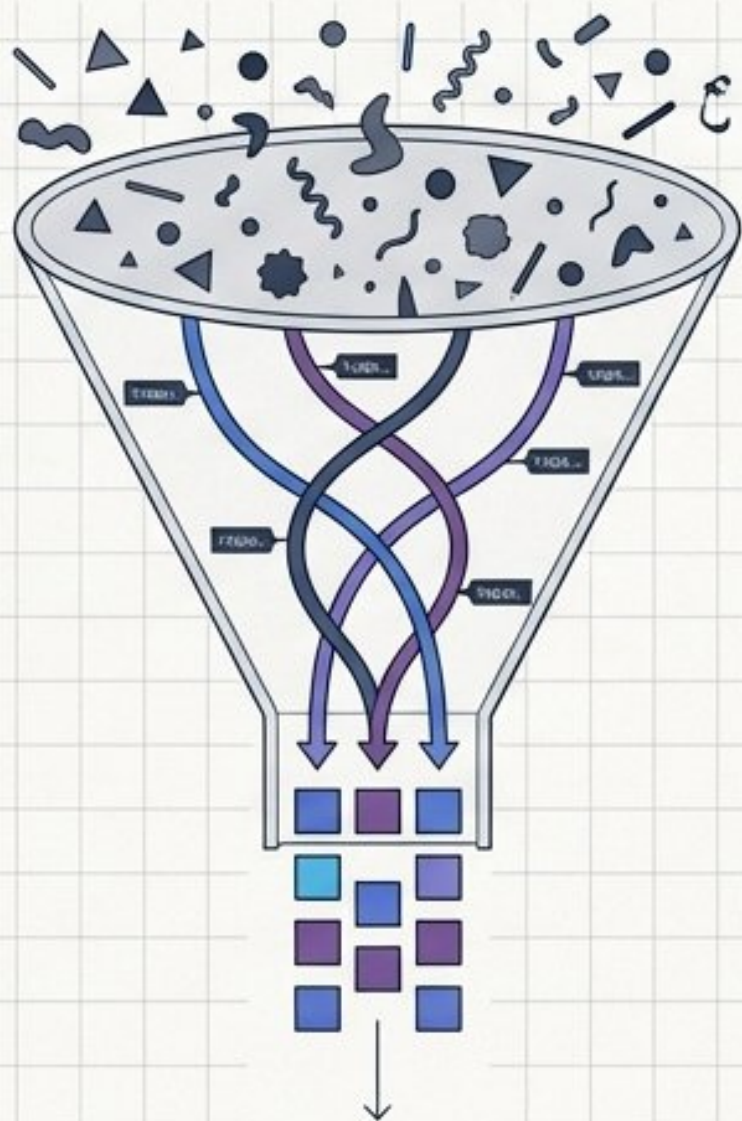
The AMACE Architecture



Autonomous Ingestion & The Universal Anomaly Format

SSIE

Entropy-based field classification.
Zero human schema mapping.



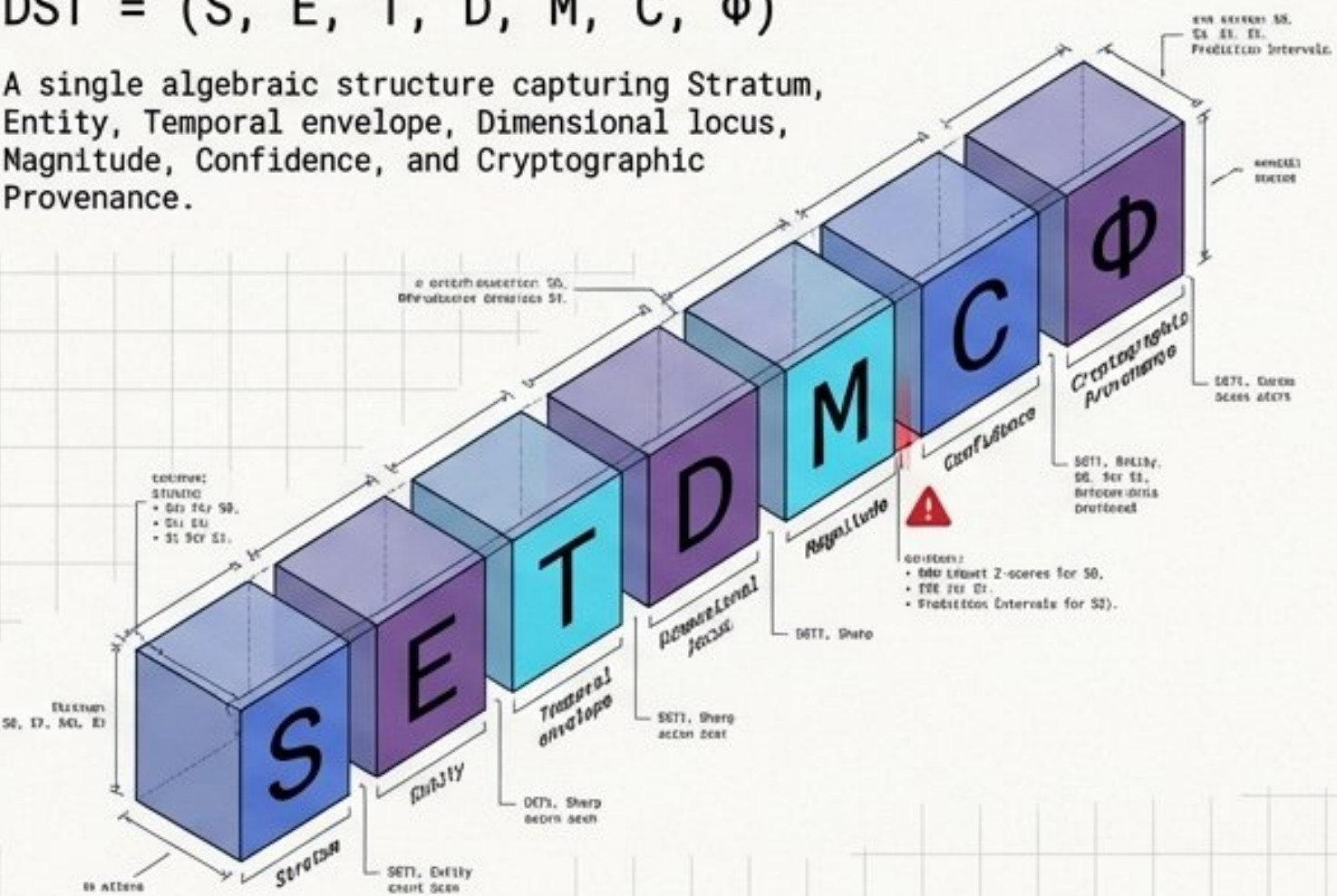
SADD

Stratum-adaptive detection (MAD-robust Z-scores for S0, PSI for S1, Prediction Intervals for S2).

The Deviation Signature Tensor (DST)

$$\text{DST} = (\text{S}, \text{E}, \text{T}, \text{D}, \text{M}, \text{C}, \Phi)$$

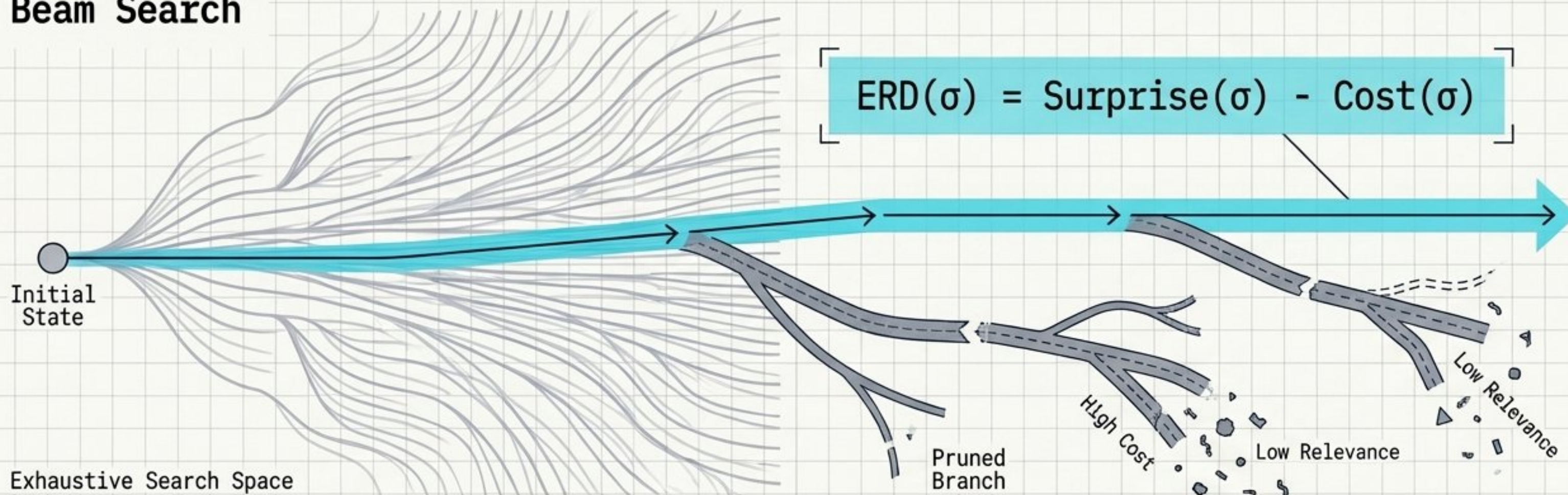
A single algebraic structure capturing Stratum, Entity, Temporal envelope, Dimensional locus, Magnitude, Confidence, and Cryptographic Provenance.



Entropic Relevance Decomposition

$$\text{ERD}(\sigma) = \text{Surprise}(\sigma) - \text{Cost}(\sigma)$$

Beam Search



The Goal

Locate the precise dimensional slice where an anomaly is concentrated.

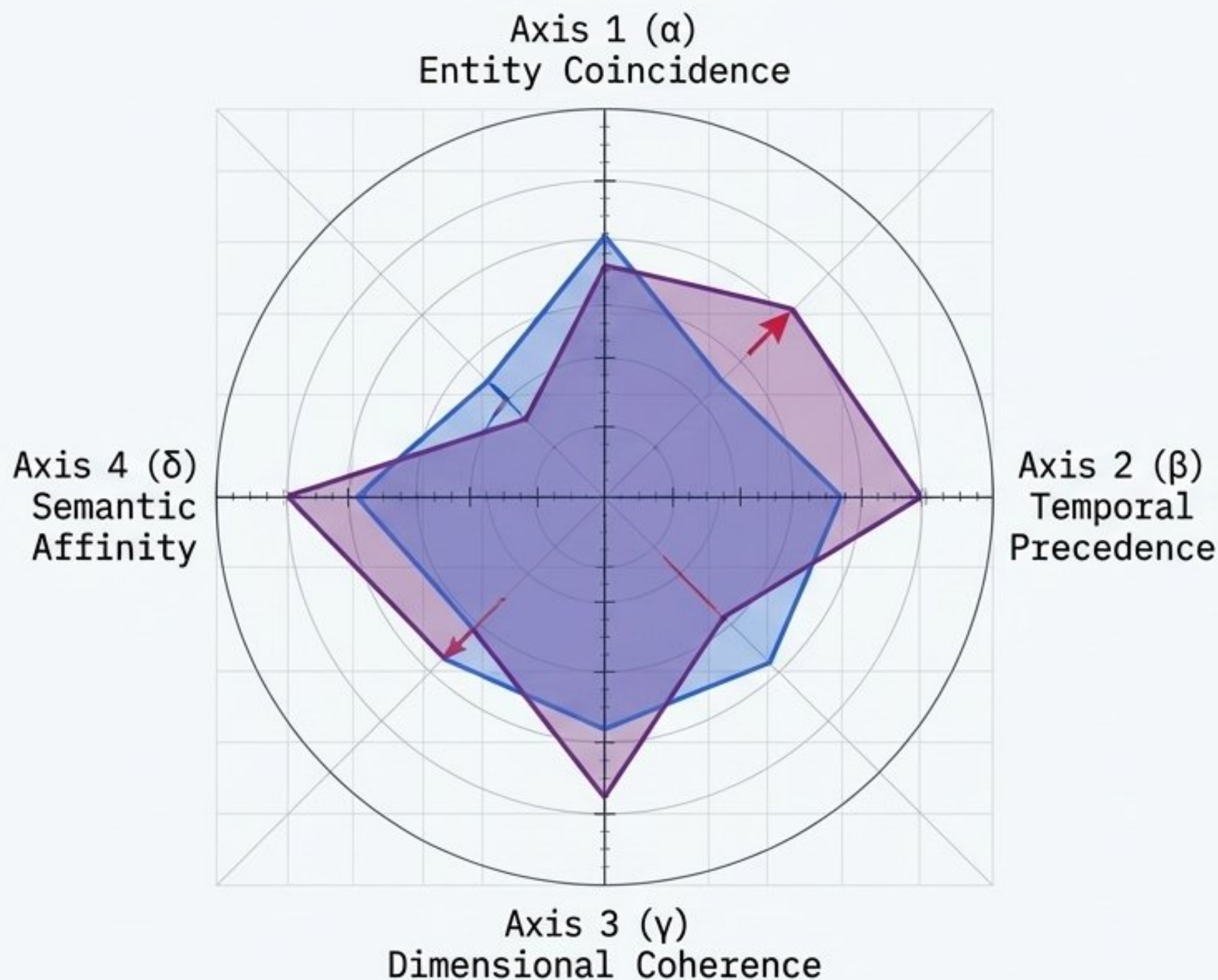
The Math

Balances information gained against the complexity of the description.

The Technical Improvement

Reduces computational complexity from exhaustive $O(K^D)$ search to $O(w \times D \times K_{\max})$, eliminating 99.94% of irrelevant dimensional combinations.

The Cross-Stratum Causal Tensor



Axis 1: **Entity Coincidence (α)**
Surprise-adjusted Jaccard similarity.

Axis 2: **Temporal Precedence (β)**
Learned stratum-to-stratum
propagation delay models.

Axis 3: **Dimensional Coherence (γ)**
Hierarchical matching (e.g.,
California maps to *San Francisco*).

Axis 4: **Semantic Affinity (δ)**
Self-trained contrastive embeddings
linking metric behaviors.

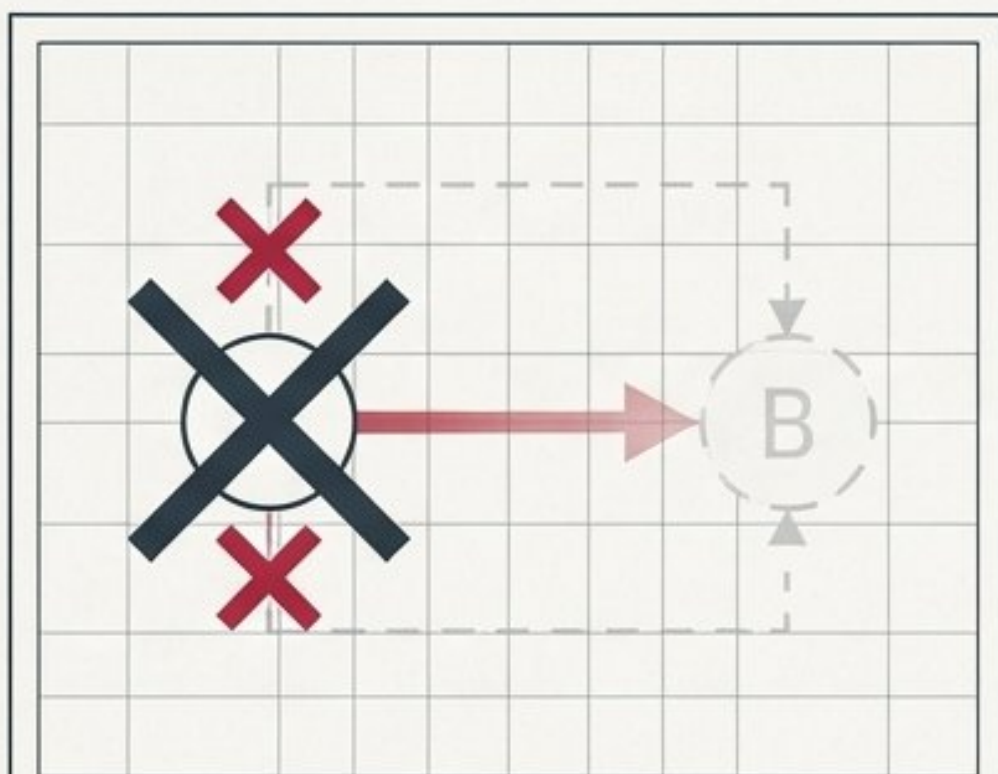
Spectral Normalization
Dynamically adapts to sudden spikes vs.
gradual drift without fixed weights.

Computational Ablation Protocol

Moving from Correlation to Provable Causation

1. Exclusivity Test

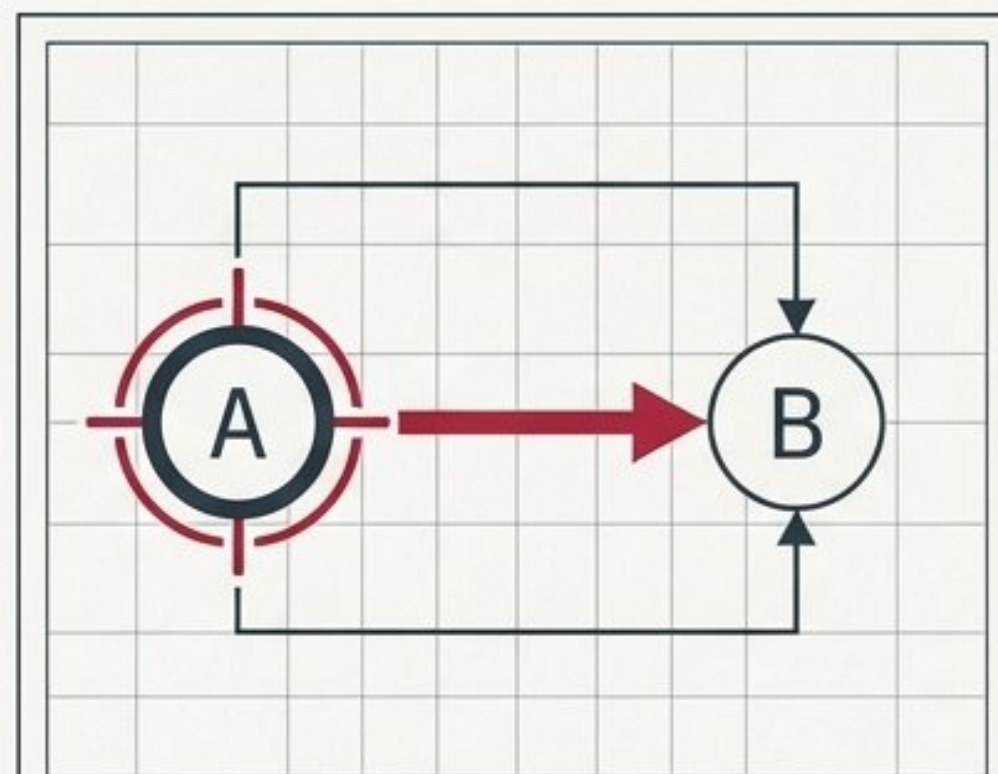
Remove anomaly A. Does anomaly B diminish? (Threshold: > 0.3 drop).



Exclusivity

2. Sufficiency Test

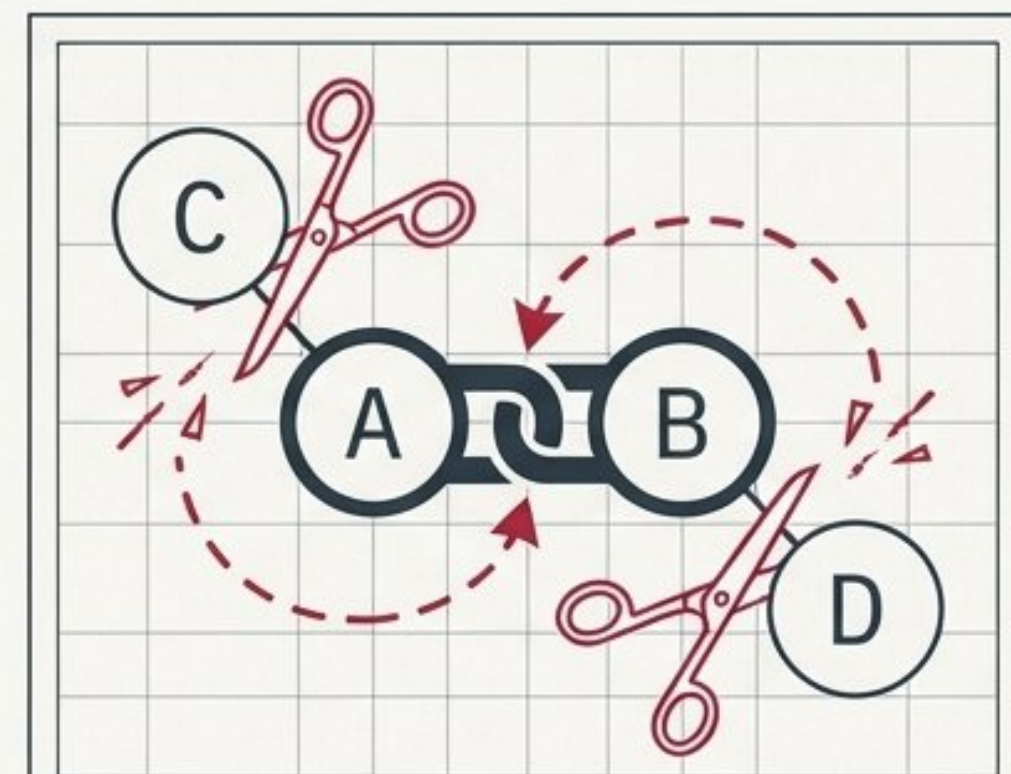
Can anomaly A alone predict B's magnitude?



Sufficiency

3. Minimality Test

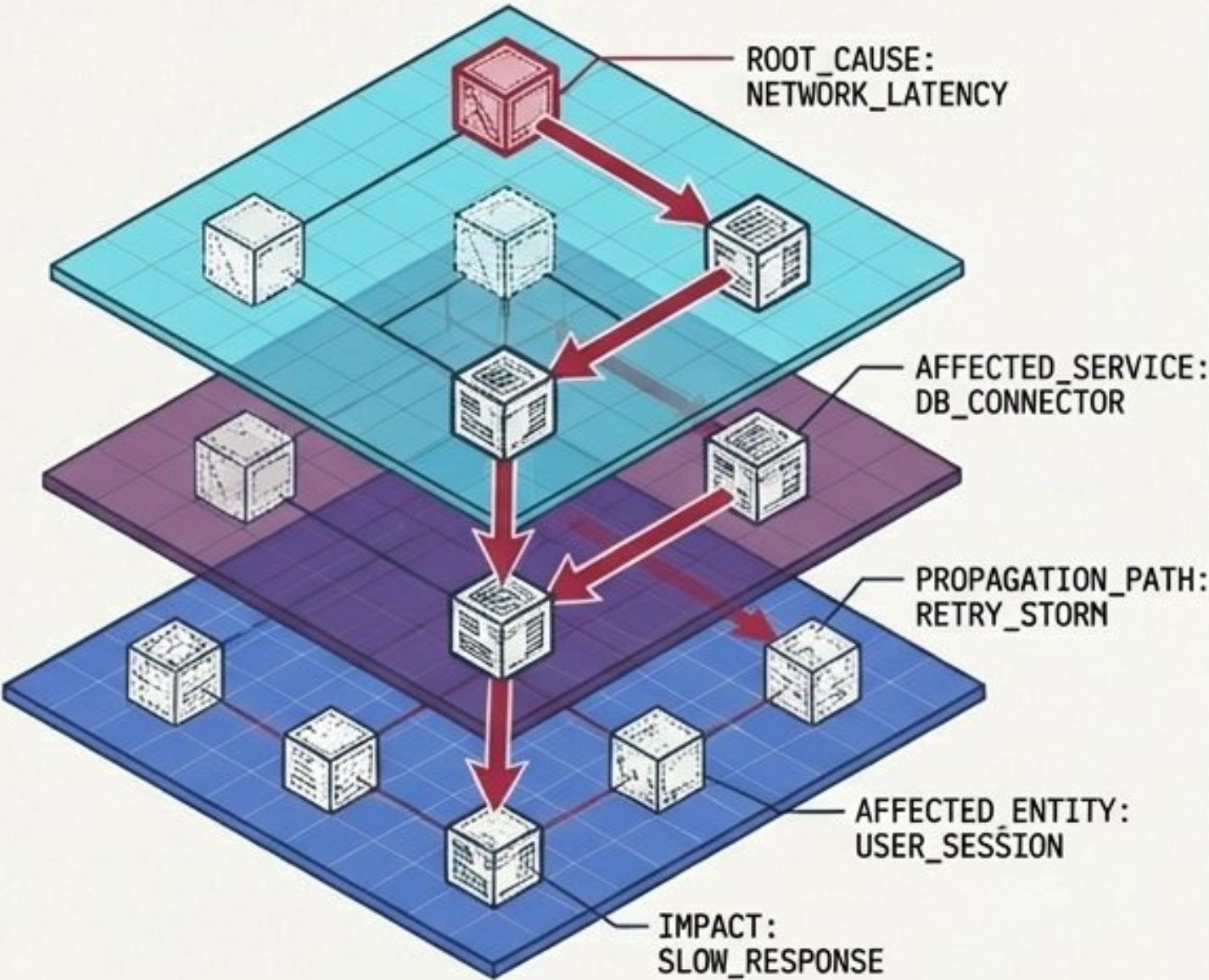
Iteratively prune multi-event correlations to the minimum sufficient set.



Minimality

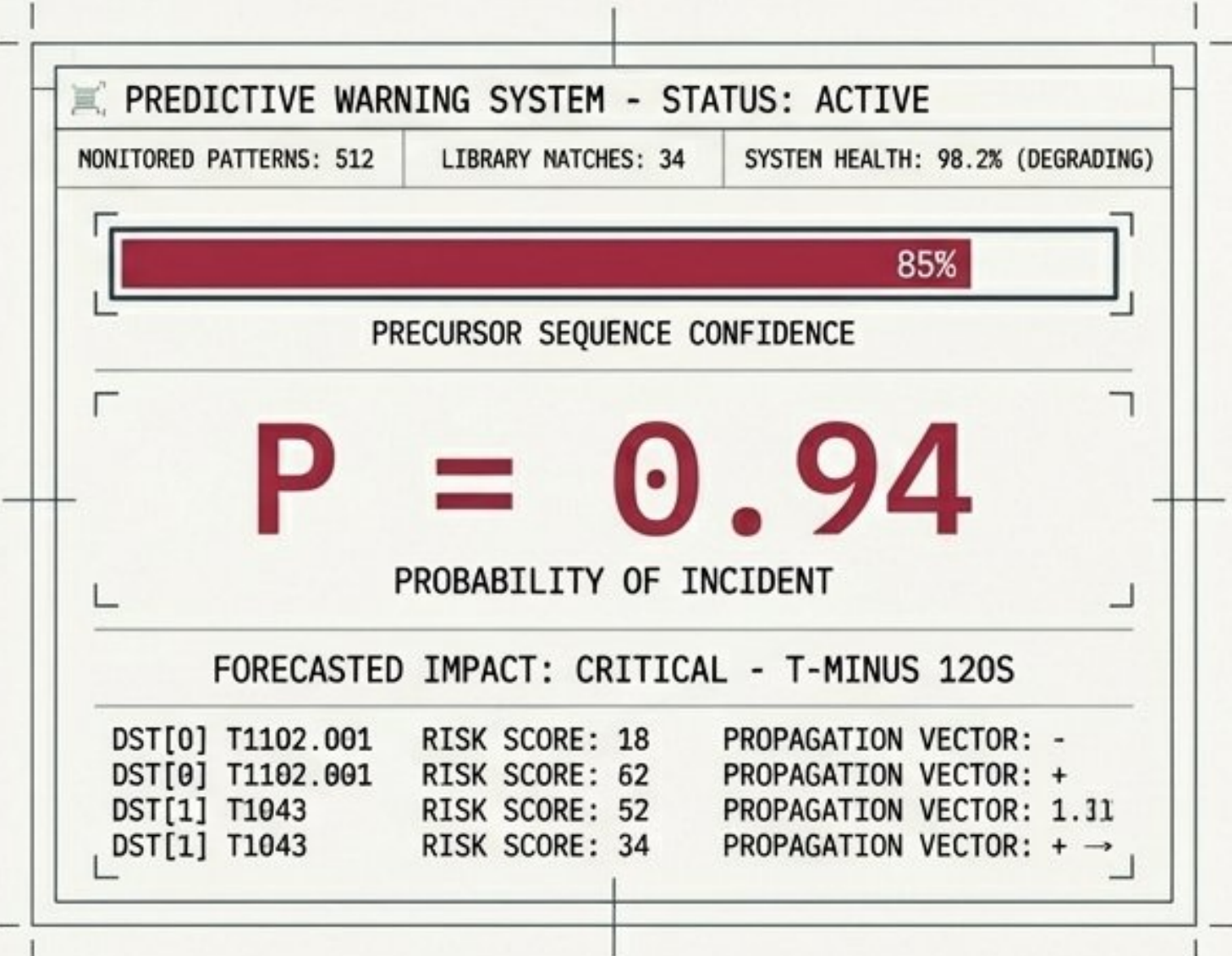
CIGS (Output)

Assembles validated DST correlations into a structured DAG identifying the root cause, propagation paths, and affected entities.



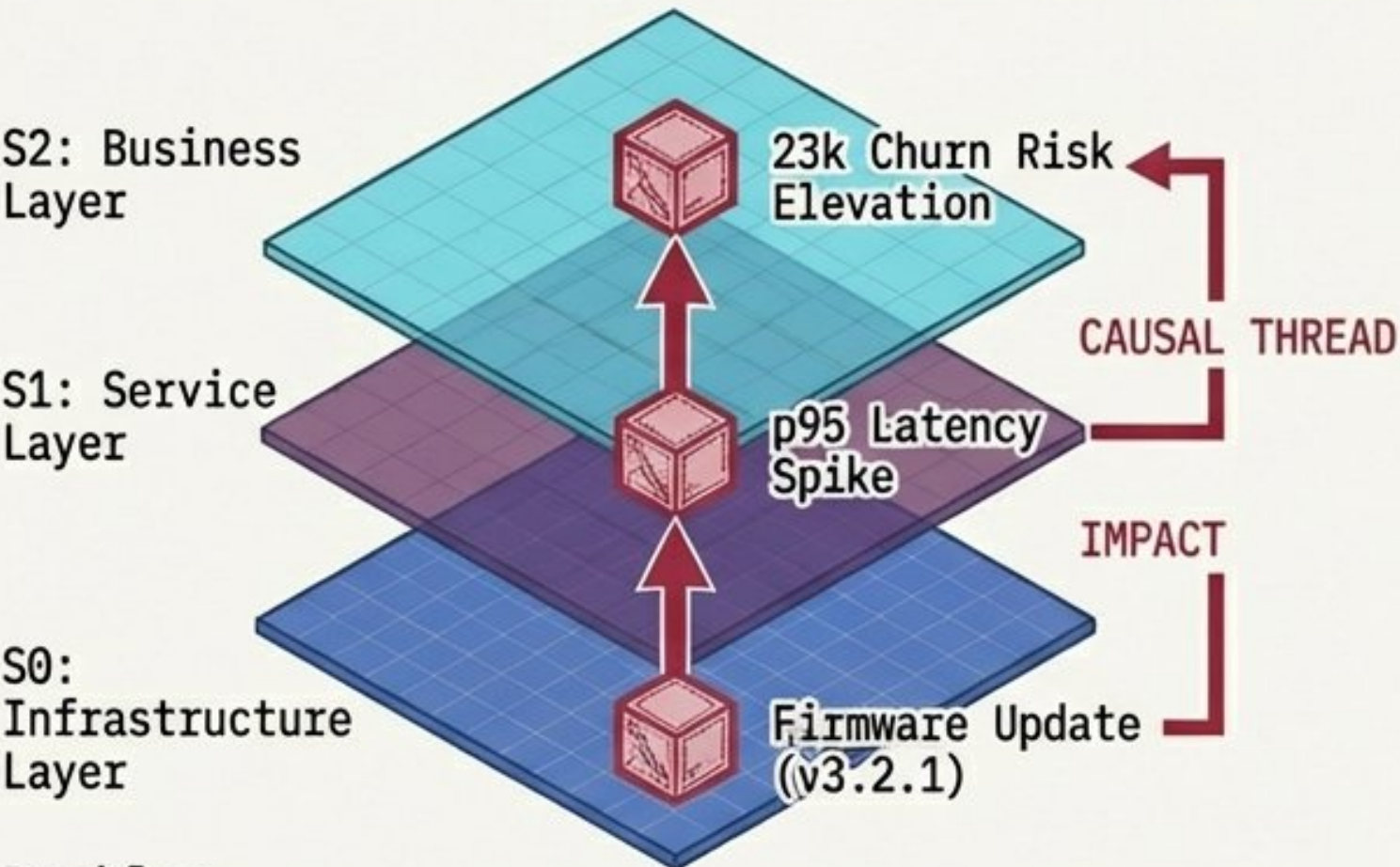
PPE (Prediction)

Matches new DST patterns against a Precursor Sequence Library. Computes $P(\text{incident} \mid \text{prefix})$ to issue predictive warnings before a systemic failure fully manifests.



Use Case 1: Telecom Cascading Service Degradation

Diagnostic Dashboard



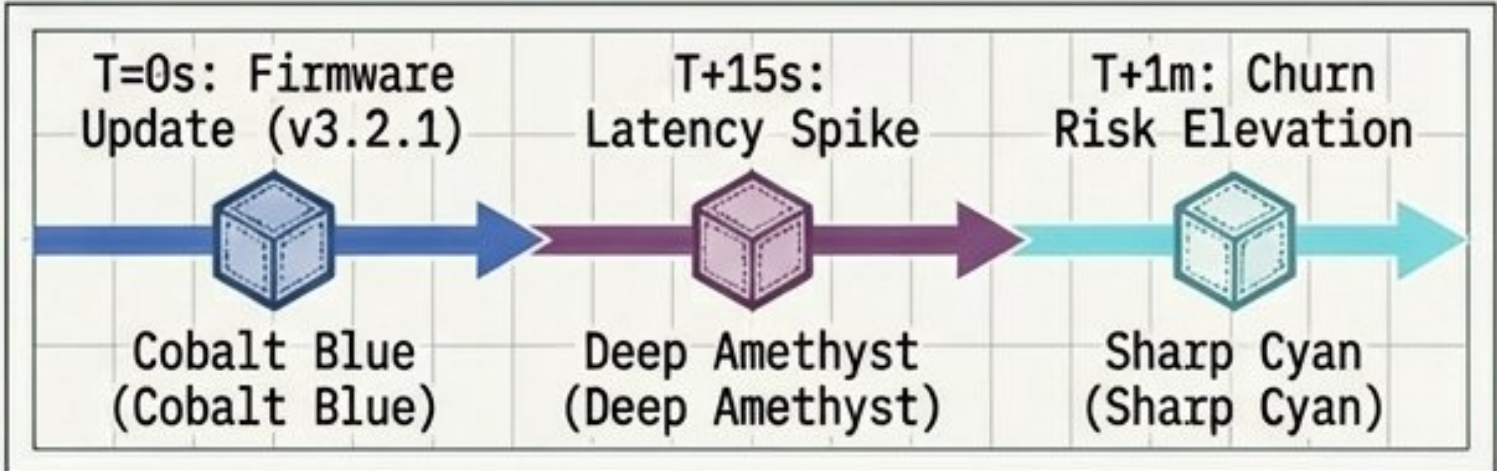
Problem

340% increase in video buffering complaints across Chicago.

AMACE Insight

Correlated an S0 edge router firmware update (v3.2.1) to an S1 p95 latency spike, directly driving an S2 churn risk elevation for 23,000 subscribers.

Sequential Event Timeline

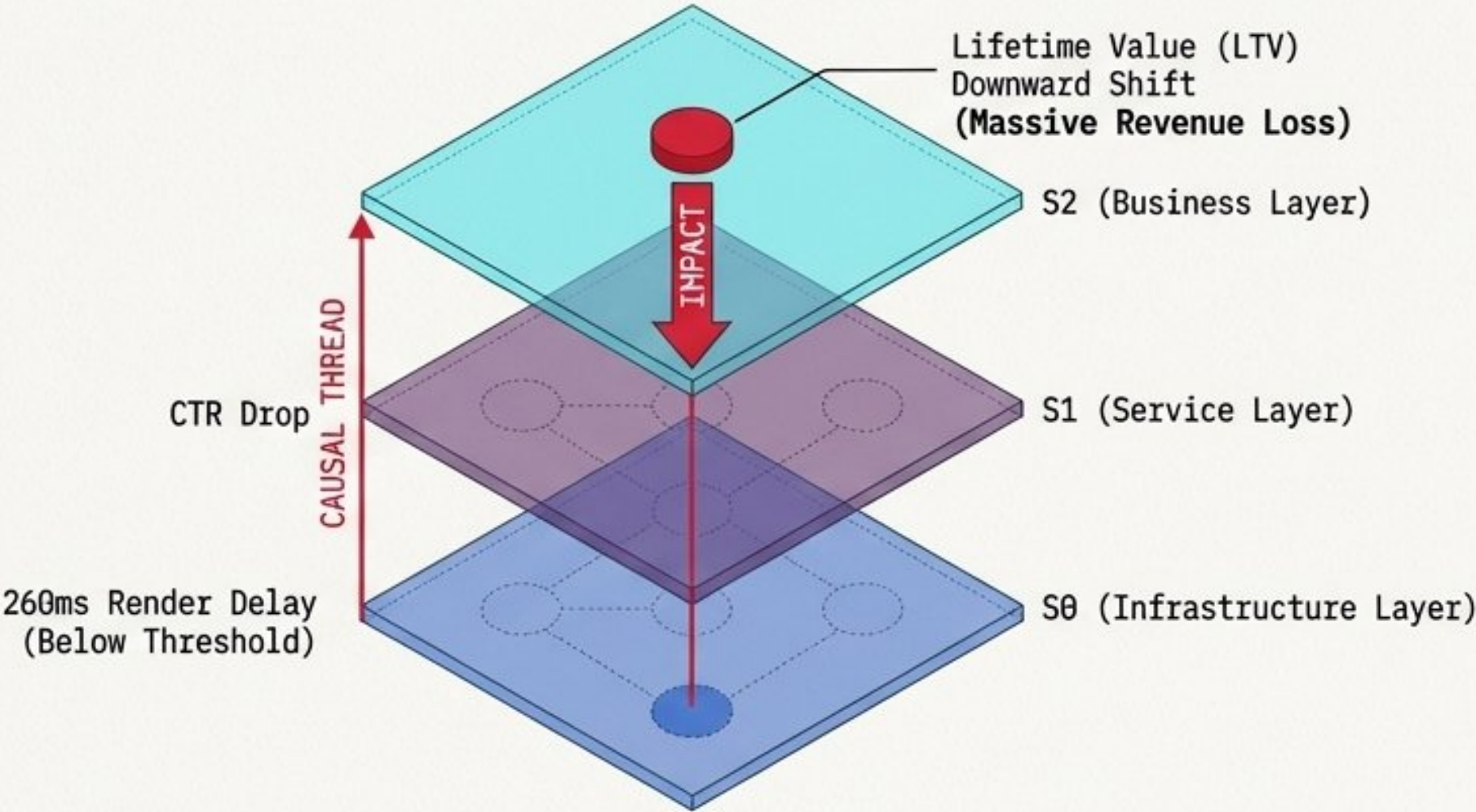


ROI

✗ Without AMACE	With AMACE ✓
14 hours manual log hunting across 3 isolated teams.	4.2 minutes to root cause. Zero configuration required.

Use Case 2: Silent Revenue Loss in Mobile Apps

Diagnostic Dashboard



Problem

Daily revenue dropped \$2.4M, but all monitoring dashboards showed green.

AMACE Insight

SADD detected a 260ms render delay in S0 (below outage thresholds). CSCT proved this delay caused a CTR drop in S1, perfectly aligning with a Lifetime Value (LTV) downward shift in S2.

Sequential Event Timeline

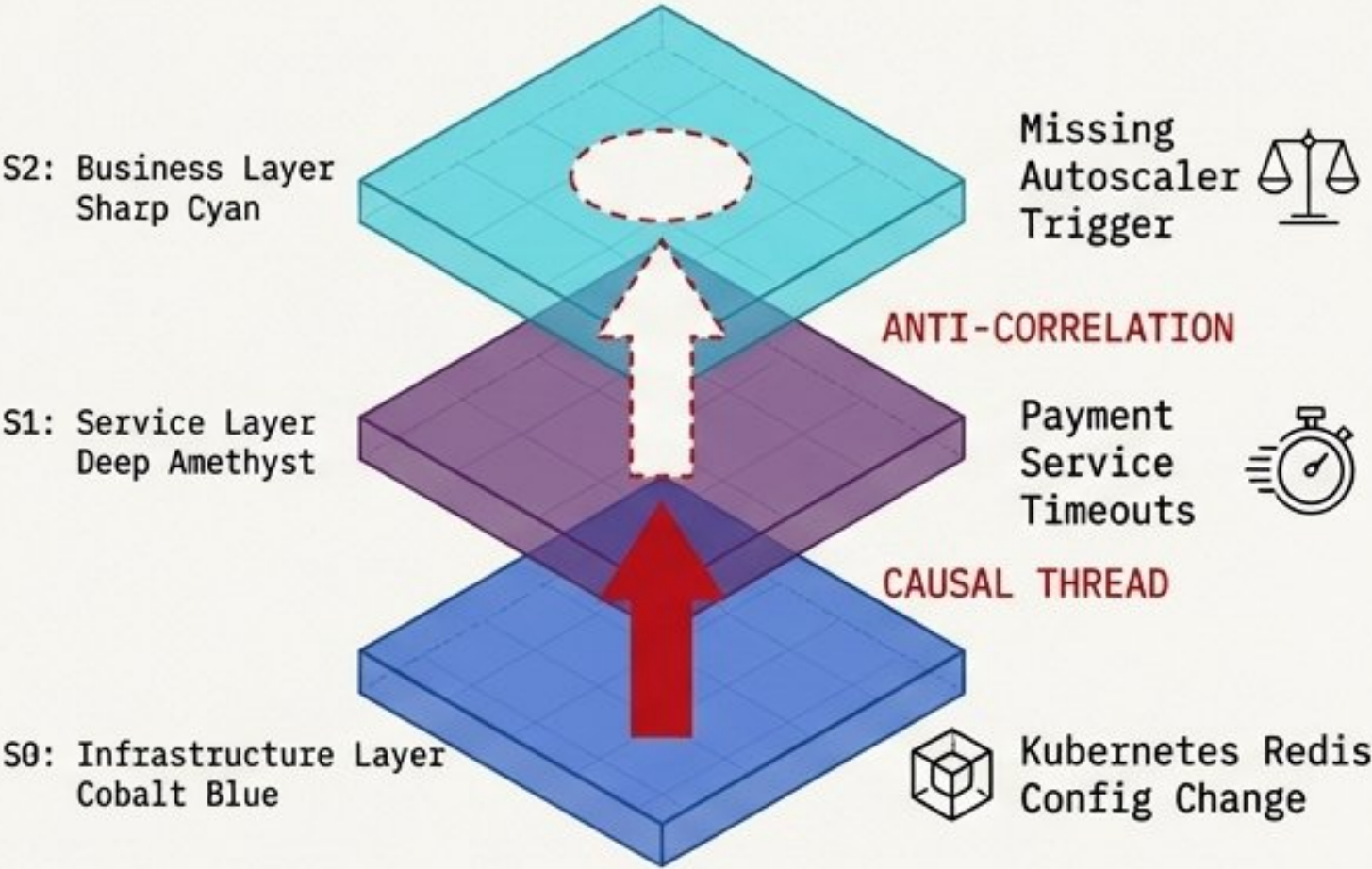


ROI

✗ Without AMACE	✓ With AMACE
Invisible to standard tools	\$7.2M saved by catching sub-threshold causal chain in 12 minutes

Use Case 3: Cloud Infrastructure & Anti-Correlation

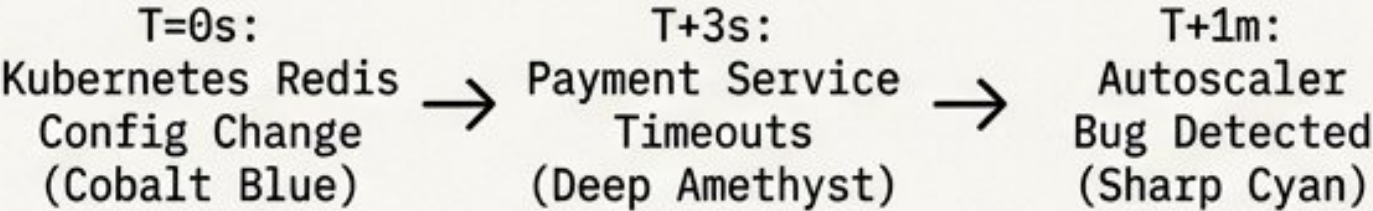
Diagnostic Dashboard



Problem: A Kubernetes Redis config change caused payment service timeouts.

AMACE Insight: Identified the Redis memory leak root cause. Crucially, the Predictive Precursor Engine detected an Anti-Correlation: the S2 autoscaler should have triggered, but didn't. This revealed a latent, secondary infrastructure bug.

Sequential Event Timeline

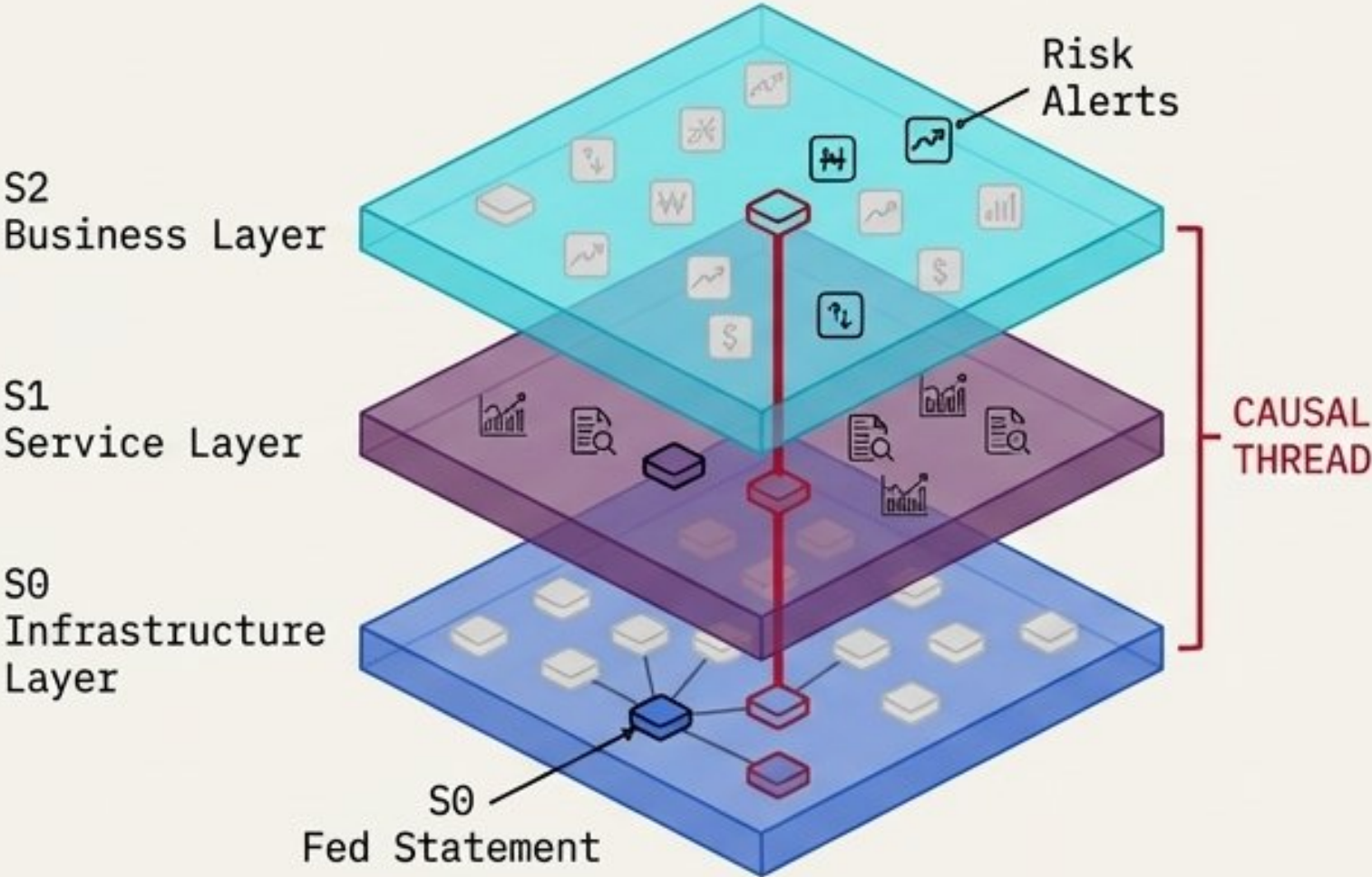


ROI

Without AMACE ✕	With AMACE ✓
6 hours to find Redis leak; autoscaler bug remains unknown.	8 minutes to dual-insight.

Use Case 4: Cross-Asset Systemic Risk in Finance

Diagnostic Dashboard



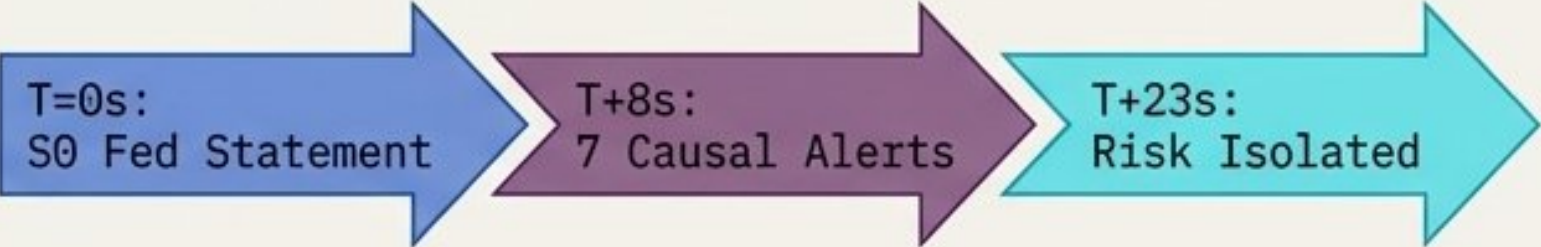
Problem

Volatile market session generates 47 independent risk alerts. Are they correlated or coincidental?

AMACE Insight

CSCT combined with Computational Ablation proved that only 7 alerts were causally chained to an S0 Fed Statement. The remaining 40 were coincidental threshold breaches.

Sequential Event Timeline

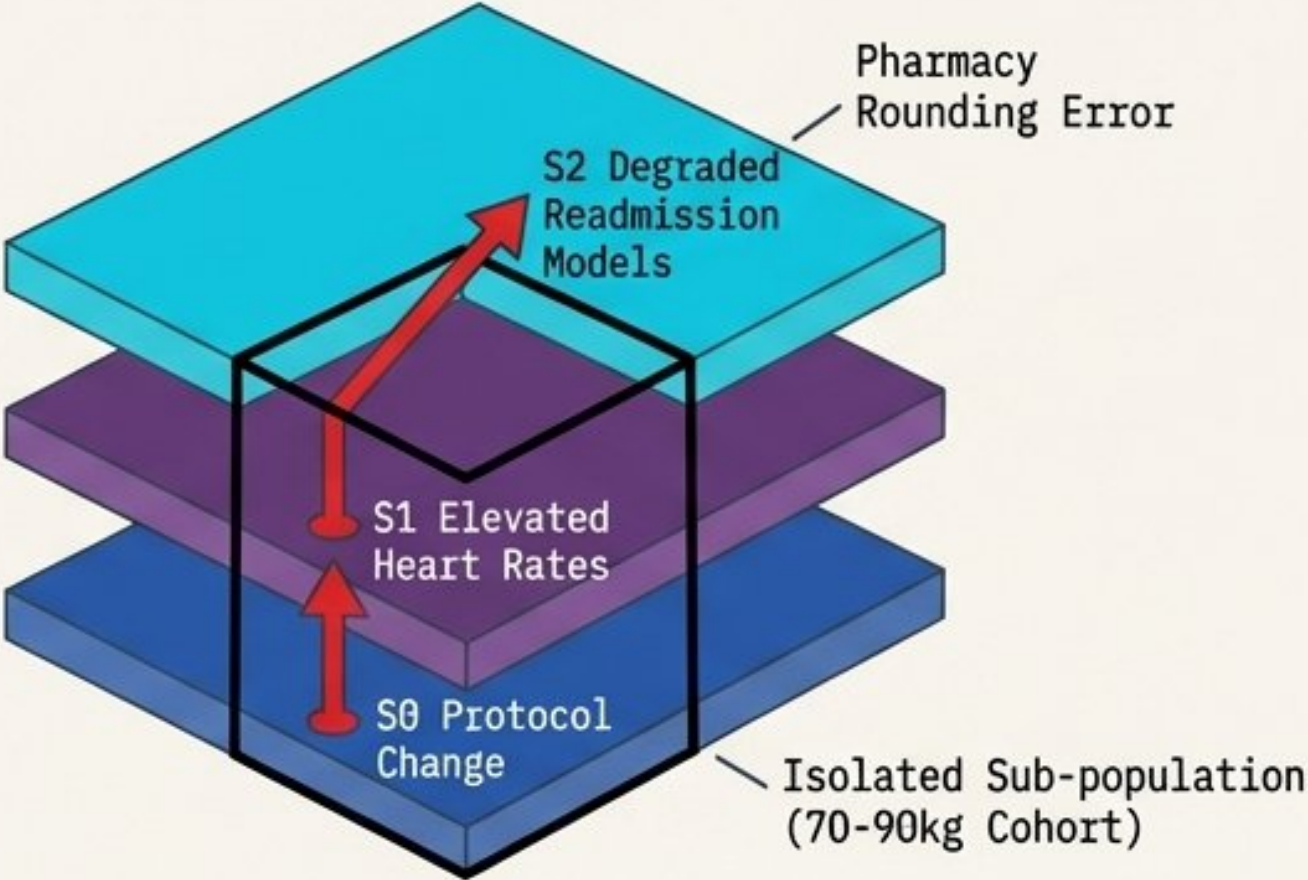


ROI

Without AMACE	With AMACE
✗ Blind liquidation of 47 positions.	✓ \$4.7M in unnecessary trading costs prevented in 23 seconds.

Use Case 5: Healthcare Clinical Adverse Events

Diagnostic Dashboard



Problem:
Unrelated alerts across pharmacy, nursing, and data science departments.

AMACE Insight:
ERD localized an anomaly to a precise sub-population (weight range 70-90kg). Correlated an S0 protocol change to S1 elevated heart rates and S2 degraded readmission models, discovering a pharmacy rounding error.

Sequential Event Timeline

T=0s: S0 Protocol Change (Cobalt Blue) → T+6h: S1 Elevated Heart Rates (Deep Amethyst) → T+48h: S2 Readmission Model Degradation (Sharp Cyan)

ROI

Without AMACE



11 days manual discovery by a pharmacy resident

With AMACE



48 hours to insight, protecting 847 patients from sub-therapeutic doses

Ground-Up Re-Engineering for IP Defensibility

Category	Version 1	AMACE v2
Terminology	Generic Signals	Formal Stratum-Specific Deviation Signatures
Data Structure	Ad-hoc JSON	7-dimensional DST tuple with algebraic operations
Search Space	Conceptual pruning	ERD with formal $O(k \cdot \log(D))$ bound proof
Correlation	Weighted sums	4-Axis Tensor with spectral normalization
Validation	Narrative counterfactuals	3-Test Computational Ablation Protocol

Surviving §101 (Alice) & §103 (Obviousness)



Attack Vector



AMACE Defense



Abstract Idea /
Just correlating data

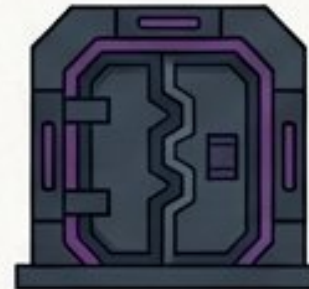


Uses **concrete data structures (DST)** and achieves **measurable technical improvement** (exponential reduction in search space).

DST
reduction:
 $O(k \cdot \log(D))$
exponential reduction



Generic computing
functions



Specific, non-generic algorithms (MDL-scored beam search, MAD-robust Z-scores).

MDL-scored beam
search, MAD-
robust Z-scores



Obvious combination
of known tools



The **\$50B monitoring industry** provides massive **teaching-away evidence** by universally **enforcing stratum isolation**. AMACE produces **unpredictable results** (e.g., **Anti-correlation**).

\$50B monitoring
industry
teaching-away
Anti-correlation

The Enforceability Map

Every internal module is engineered to produce an externally observable artifact.

Claim 1b (SSIE)

Observed via zero-configuration ingestion behavior.

Claim 1c (SADD)

DST format inherently visible in structured API responses.

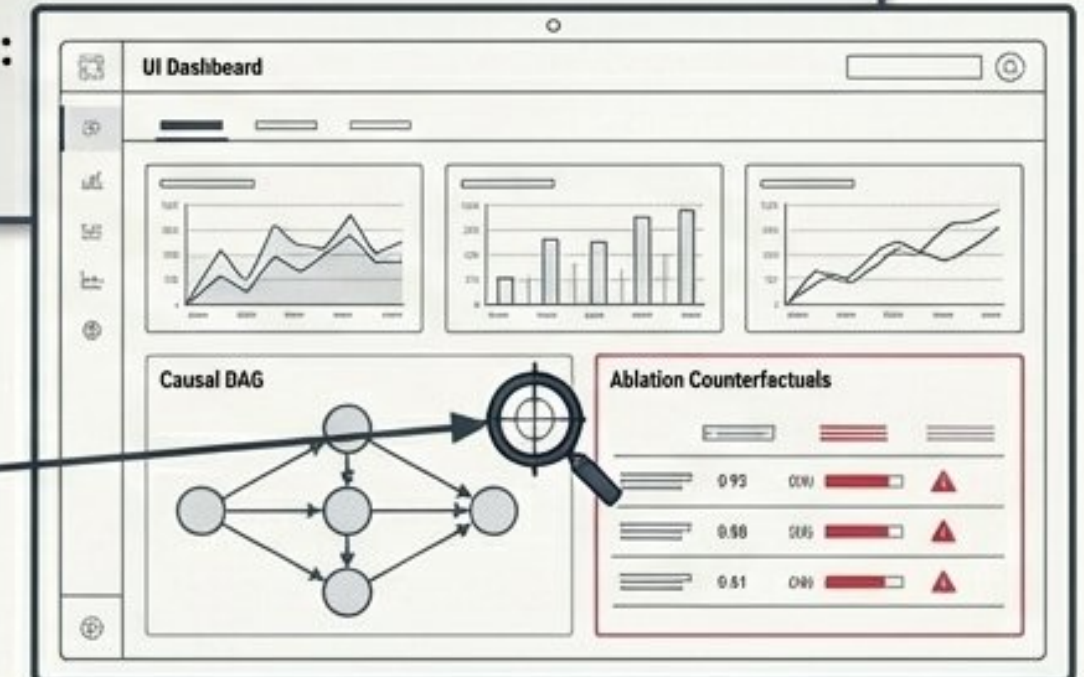
Claim 1e (CSCT)

4-axis similarity scores exposed in correlation output arrays.

Claim 1f & 9 (CAP & UI)

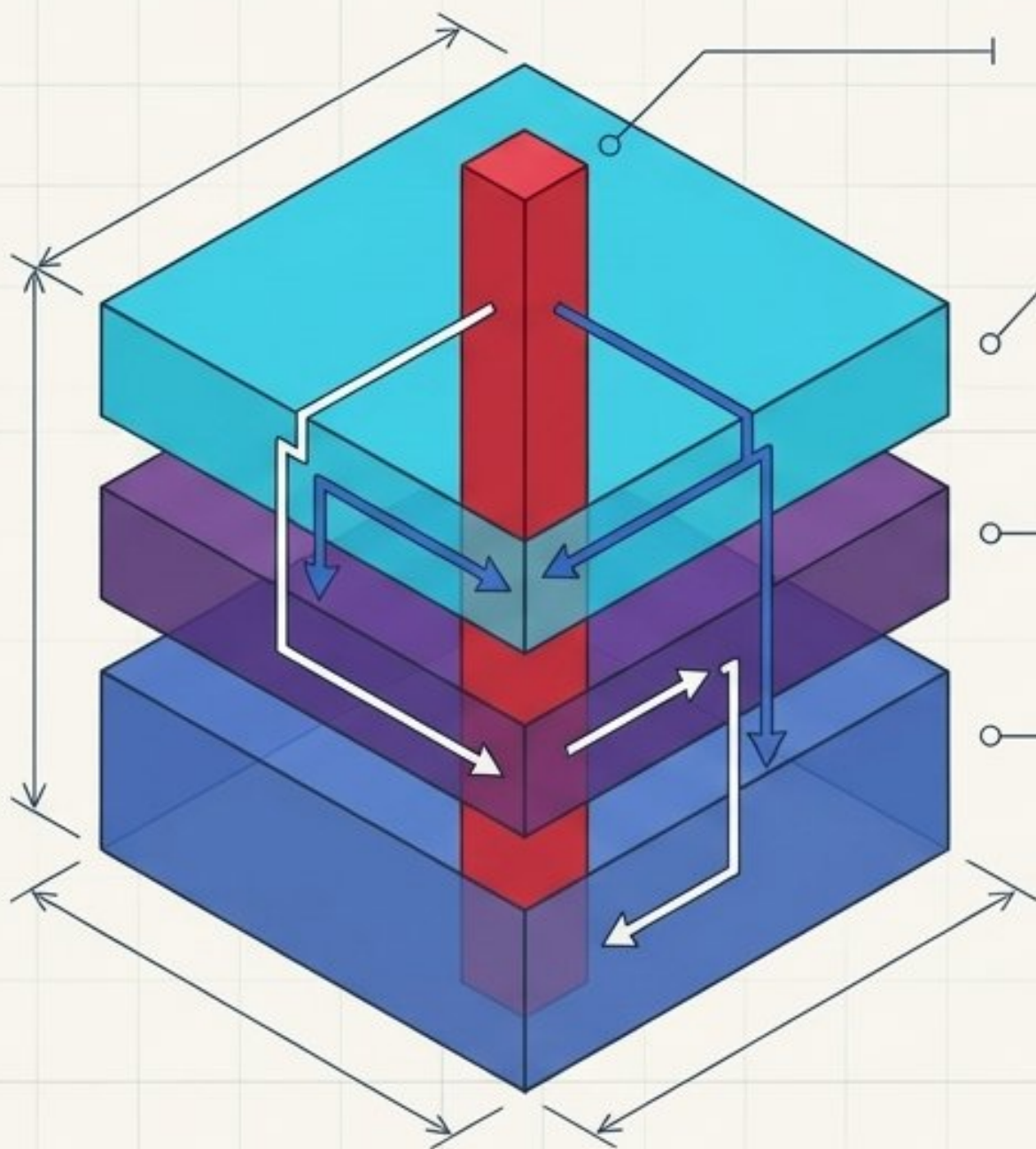
Ablation counterfactuals and Causal DAG explicitly rendered in the competitor's UI.

```
{
  "ingestion_behavior": "zero_configuration",
  "dst_format": {
    "tuple_7d": [0.1, 0.5, 0.8, 0.3, 0.9, 0.2, 0.6],
    "algebraic_ops": true
  },
  "correlation_output": {
    "similarity_scores": {
      "axis_1": 0.92,
      "axis_2": 0.88,
      "axis_3": 0.75,
      "axis_4": 0.61
    }
  },
  "spectral_norm":
}
```



The AMACE Advantage

Algorithmic Superiority



Algorithmic Superiority:

Solves a mathematically intractable problem via $O(k \cdot \log(D))$ optimization and counterfactual validation.

Provable Multi-Domain ROI:

Generates immediate, un-configured value in telecommunications, cloud infrastructure, healthcare, and finance.

Bulletproof IP Moat:

Protected by rigorous mathematical claims, observable enforcement vectors, and undeniable technical improvement to the computer system itself.